[19]
5s

```python
# installer roboflow og importer datasæt fra roboflow

!pip install roboflow

from roboflow import Roboflow


rf = Roboflow(api_key="[redacted]")
project = rf.workspace("valdemars-workspace").project("mandatory-1-ml")


# korrekt version med 'person' objects.
version = project.version(4)
dataset = version.download("yolov11")
```

```
Requirement already satisfied: roboflow in /usr/local/lib/python3.12/dist-packages (1.2.16)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from roboflow) (2026.2.25)
Requirement already satisfied: idna==3.7 in /usr/local/lib/python3.12/dist-packages (from roboflow) (3.7)
Requirement already satisfied: cycler in /usr/local/lib/python3.12/dist-packages (from roboflow) (0.12.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from roboflow) (1.4.9)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from roboflow) (3.10.0)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from roboflow) (2.0.2)
Requirement already satisfied: opencv-python-headless==4.10.0.84 in /usr/local/lib/python3.12/dist-packages (from roboflow) (4.10.0.84)
Requirement already satisfied: Pillow>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from roboflow) (11.3.0)
Requirement already satisfied: pillow-avif-plugin<2 in /usr/local/lib/python3.12/dist-packages (from roboflow) (1.5.5)
Requirement already satisfied: python-dateutil in /usr/local/lib/python3.12/dist-packages (from roboflow) (2.9.0.post0)
Requirement already satisfied: python-dotenv in /usr/local/lib/python3.12/dist-packages (from roboflow) (1.2.1)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from roboflow) (2.32.4)
Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages (from roboflow) (1.17.0)
Requirement already satisfied: urllib3>=1.26.6 in /usr/local/lib/python3.12/dist-packages (from roboflow) (2.5.0)
Requirement already satisfied: tqdm>=4.41.0 in /usr/local/lib/python3.12/dist-packages (from roboflow) (4.67.3)
Requirement already satisfied: PyYAML>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from roboflow) (6.0.3)
Requirement already satisfied: requests-toolbelt in /usr/local/lib/python3.12/dist-packages (from roboflow) (1.0.0)
Requirement already satisfied: filetype in /usr/local/lib/python3.12/dist-packages (from roboflow) (1.2.0)
Requirement already satisfied: pi-heif<2 in /usr/local/lib/python3.12/dist-packages (from roboflow) (1.3.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->roboflow) (1.3.3)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->roboflow) (4.61.1)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->roboflow) (26.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->roboflow) (3.3.2)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->roboflow) (3.4.4)

loading Roboflow workspace...
loading Roboflow project...
```

```python
# træn YOLOv11 model

!pip install ultralytics

from ultralytics import YOLO

model = YOLO("yolo11n.pt")

model.train(
    data=dataset.location + "/data.yaml",
    # først sat op til 50 fra 25, precision var for lav ved 25 epochs (~.22)
    # herefter sat op til 200, da p stadig var lidt for lavt (~.49)
    epochs=200,
    imgsz=640
)
```

```
        0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,
0.6,     0.6,
        0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,
0.6,     0.6,
        0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,
0.6,     0.6,
        0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,
0.6,     0.6,
        0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,
0.6,     0.6,
        0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,
0.6,     0.6,
        0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6,     0.6, 0.59818, 0.59549, 0.59279, 0.59009, 0.58739,  0.5847,   0.582,  0.5793,  0.5766, 0.57391, 0.57121, 0.56851,
0.56581, 0.56312,
    0.56042, 0.55772, 0.55502, 0.55233, 0.54963, 0.54693, 0.54424, 0.54154, 0.53884, 0.53614, 0.53345, 0.53075, 0.52805, 0.52535, 0.52266, 0.51996, 0.51726, 0.51456, 0.51187, 0.50917, 0.50647,
0.50377, 0.50108,
        0.5,     0.5,     0.5, 0.46544, 0.41419, 0.39663, 0.39197, 0.38731, 0.38265, 0.37799, 0.37333, 0.36868, 0.36402, 0.35936,  0.3547, 0.35004, 0.34538, 0.34072, 0.33606,  0.3314, 0.32674,
0.32208, 0.31742,
    0.31277, 0.30811, 0.30345,  0.2997, 0.29853, 0.29737,  0.2962, 0.29504, 0.29387, 0.29271, 0.29154, 0.29038, 0.28921, 0.28805, 0.28688, 0.28572, 0.28455, 0.28339, 0.28222, 0.28106,  0.2799,
0.27873, 0.27757,
     0.2764, 0.27524, 0.27407, 0.27291, 0.27174, 0.27058, 0.26941, 0.26825, 0.26708, 0.26592, 0.26475, 0.26359, 0.26242, 0.26126, 0.26009, 0.25893, 0.25776,  0.2566, 0.25543, 0.25427,  0.2531,
0.25194, 0.25078,
    0.24961, 0.24845, 0.24728, 0.24612, 0.24495, 0.24379, 0.24262, 0.24146, 0.24029, 0.23913, 0.23796,  0.2368, 0.23563, 0.23447,  0.2333, 0.23214, 0.23097, 0.22981, 0.22864, 0.22748, 0.22631,
0.22515, 0.22398,
    0.22282, 0.22166, 0.22049, 0.21933, 0.21816,   0.217, 0.21583, 0.21467,  0.2135, 0.21234, 0.21117, 0.21001, 0.20884, 0.20768, 0.20651, 0.20535, 0.20418, 0.20302, 0.20185, 0.20069, 0.19973,
0.19908, 0.19843,
    0.19777, 0.19712, 0.19647, 0.19582, 0.19516, 0.19451, 0.19386,  0.1932, 0.19255,  0.1919, 0.19125, 0.19059, 0.18994, 0.18929, 0.18863, 0.18798, 0.18733, 0.18668, 0.18602, 0.18537, 0.18472,
0.18406, 0.18341,
    0.18276, 0.18211, 0.18145,  0.1808, 0.18015, 0.17949, 0.17884, 0.17819, 0.17754, 0.17688, 0.17623, 0.17558, 0.17492, 0.17427, 0.17362, 0.17297, 0.17231, 0.17166, 0.17101, 0.17035,  0.1697,
0.16905, 0.16839,
    0.16774, 0.16709, 0.16644, 0.16578, 0.16513, 0.16448, 0.16382, 0.16317, 0.16252, 0.16187, 0.16121, 0.16056, 0.15991, 0.15925,  0.1586, 0.15795,  0.1573, 0.15664, 0.15599, 0.15534, 0.15468,
0.15403, 0.15338,
    0.15273, 0.15207, 0.15142, 0.15077, 0.15011, 0.14946, 0.14881, 0.14816,  0.1475, 0.14685,  0.1462, 0.14554, 0.14489, 0.14424, 0.14359, 0.14293, 0.14228, 0.14163, 0.14097, 0.14032, 0.13967,
0.13902, 0.13836,
    0.13771, 0.13706,  0.1364, 0.13575,  0.1351, 0.13444, 0.13379, 0.13314, 0.13249, 0.13183, 0.13118, 0.13053, 0.12987, 0.12922, 0.12857, 0.12792, 0.12726, 0.12661, 0.12596,  0.1253, 0.12465,
0.124,   0.12335,
    0.12269, 0.12204, 0.12139, 0.12073, 0.12008, 0.11943, 0.11878, 0.11812, 0.11747, 0.11682, 0.11616, 0.11551, 0.11486, 0.11421, 0.11355,  0.1129, 0.11225, 0.11159, 0.11094, 0.11029, 0.10964,
0.10898, 0.10833,
    0.10768, 0.10702, 0.10637, 0.10572, 0.10507, 0.10441, 0.10376, 0.10311, 0.10245,  0.1018, 0.10115,  0.1005,       0,       0,       0,       0,       0,       0,       0,       0,       0,
0,       0,
          0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,
0,       0,
          0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,
0,       0,
          0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0,
0,       0,
          0,       0,       0,       0,       0,       0,       0,       0,       0,       0,       0]]), 'Confidence', 'Recall']]
fitness: np.float64(0.4145753696350535)
keys: ['metrics/precision(B)', 'metrics/recall(B)', 'metrics/mAP50(B)', 'metrics/mAP50-95(B)']
maps: array([    0.41458])
names: {0: 'person'}
nt_per_class: array([10])
nt_per_image: array([10])
results_dict: {'metrics/precision(B)': 0.815128103589642, 'metrics/recall(B)': 0.6, 'metrics/mAP50(B)': 0.6468026521321563, 'metrics/mAP50-95(B)': 0.4145753696350535, 'fitness': 0.4145753696350535}
save_dir: PosixPath('/content/runs/detect/train7')
speed: {'preprocess': 0.1964845999282261, 'inference': 2.260511599979509, 'loss': 0.0010149999980058055, 'postprocess': 1.2146295000093232}
stats: {'tp': [], 'conf': [], 'pred_cls': [], 'target_cls': [], 'target_img': []}
task: 'detect'
```

```python
# ved evaluering kan vi se at vi nu har en rimelig veltrænet model
results = model.val()
metrics = results.results_dict

print(f"Precision: {metrics['metrics/precision(B)']:.3f}")
print(f"Recall:    {metrics['metrics/recall(B)']:.3f}")
print(f"mAP50:     {metrics['metrics/mAP50(B)']:.3f}")
```

```
Ultralytics 8.4.21 🚀 Python-3.12.12 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
YOLO11n summary (fused): 101 layers, 2,582,347 parameters, 0 gradients, 6.3 GFLOPs
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 1257.7±330.0 MB/s, size: 45.6 KB)
val: Scanning /content/mandatory-1-ml-4/valid/labels.cache... 10 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━ 10/10 2.1Mit/s 0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━ 1/1 3.6it/s 0.3s
                   all         10         10      0.816        0.6      0.643      0.396
Speed: 1.6ms preprocess, 5.2ms inference, 0.0ms loss, 3.2ms postprocess per image
Results saved to /content/runs/detect/val3
Precision: 0.816
Recall:    0.600
mAP50:     0.643
```

Colab paid products - Cancel contracts here

Captured by FireShot Pro: 06 March 2026, 10:49:04
https://getfireshot.com